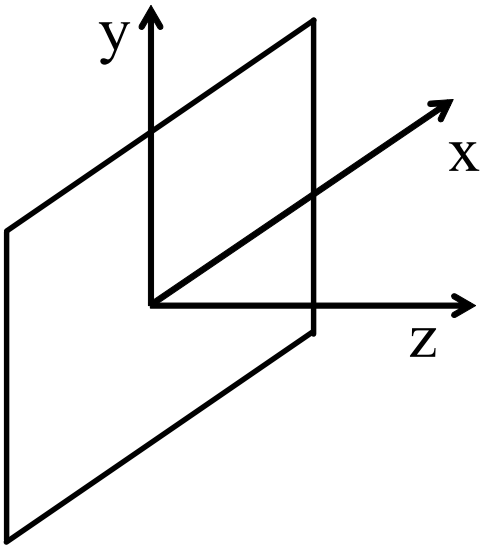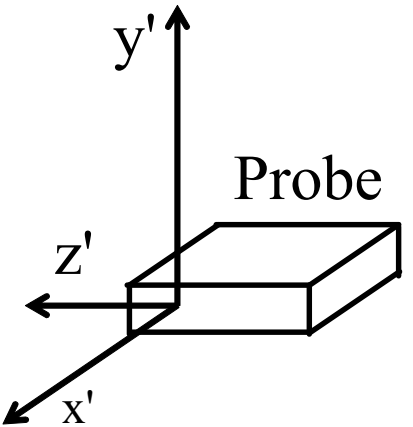Sampling plane at $z=z_0$